

Hierarchical RAG Pipeline

Child-Chunk Retrieval with Parent Expansion

RAG-Based Energy Forecasting

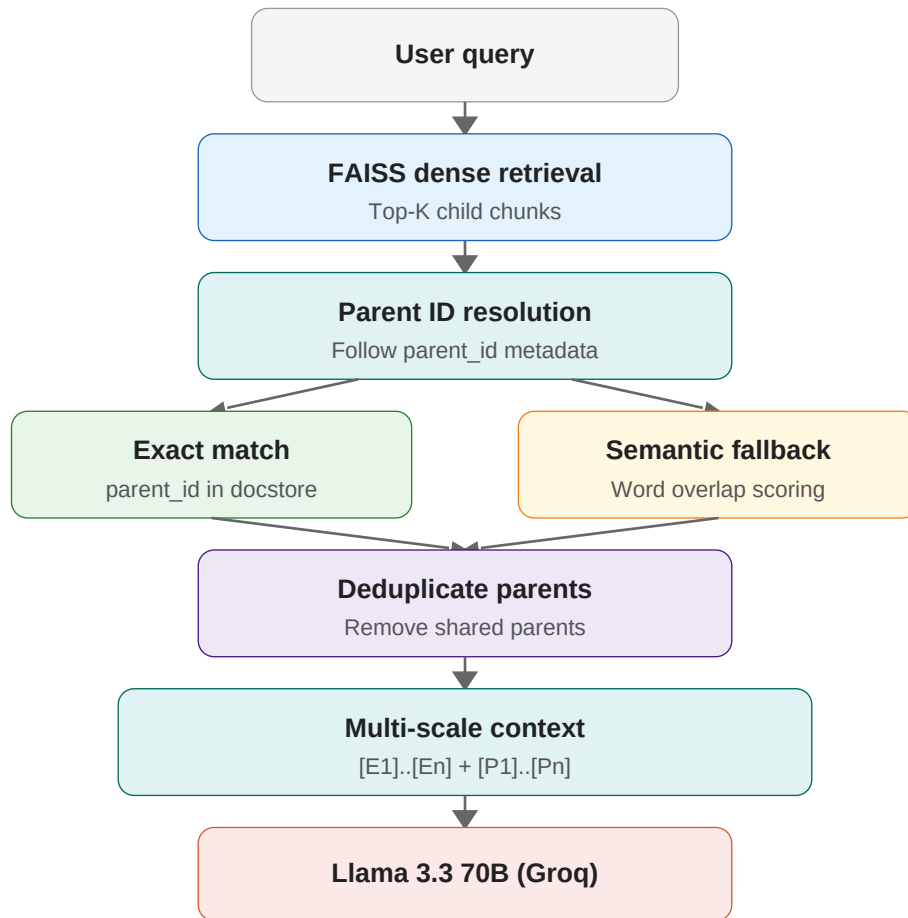
MSc Thesis — LJMU

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1. Overview

The Hierarchical RAG pipeline (EXP_04) retrieves fine-grained child chunks via FAISS dense search, then automatically expands the context by resolving each child's parent chunk at the next-higher granularity level. This provides the LLM with both specific data points (daily loads) and broader context (weekly/monthly trends), reducing hallucination by grounding answers in multi-scale evidence.

2. Architecture Diagram



3. Granularity Hierarchy

Level	Granularity	Example	Parent Level
1 (finest)	Daily	Zone 1 load on 2005-01-15	Weekly
2	Weekly	Zone 1 ISO week 3, 2005	Monthly
3	Monthly	Zone 1 January 2005	Seasonal
4 (coarsest)	Seasonal	Zone 1 Winter 2005	None

Each KB chunk stores a `parent_id` in its metadata, linking it to the next-higher granularity chunk. The hierarchical retriever follows these links to build a multi-resolution context window.

4. Parent Resolution

4.1 Exact Resolution

When a child chunk's `parent_id` matches an existing KB chunk `row_id`, the parent is resolved directly via dictionary lookup. This is the preferred path.

4.2 Semantic Fallback

When exact `parent_id` lookup fails (common when the KB sample does not contain the specific parent week/month referenced by daily chunks), the retriever falls back to `_find_parent_semantic()`. This function identifies the closest available chunk at the next-higher granularity level by computing word overlap scores between the child chunk text and all candidate parent chunks, selecting the best match.

The semantic fallback resolves 14/14 daily chunks that previously returned zero parents in the pilot KB sample.

5. Context Structure

The final context passed to the LLM contains two sections: child chunks (tagged as [E1], [E2], etc.) providing specific data points, and parent chunks (tagged as [P1], [P2], etc.) providing broader temporal context. This dual-level evidence allows the model to cite specific values while understanding trends.

6. Key Design Decisions

- Children retrieved via dense search (captures semantic intent)
- Parents resolved via metadata links (preserves hierarchical structure)
- Semantic fallback prevents zero-parent failures in sampled KBs
- Deduplication ensures no redundant parent context
- Parent docs marked with `retrieval_method='parent_expansion'` for attribution tracking

7. Key Files

File	Purpose
src/retrieval/hierarchical.py	HierarchicalRetriever + _find_parent_semantic()
experiments/exp_04_hierarchical_rag.py	Experiment orchestration
src/knowledge_base/chunk_builder.py	Enriched chunk text with semantic headers